# PAPER_2122_CONSTANT_TRIPLE_CONVERGENCE_BETA_I_RHO_VAC_C_FIRST_3_CONSTANT_INSTANCE_PAPER_2121_PREDICTION_VALIDATION_UQFF_LANDMARK

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# PAPER_2122 — Constant-Triple Convergence $\beta_i$ × $\rho_{vac}$ × c: First 3-Constant Instance in the QCalc Projection Layer (REVISED)

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**Project:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.73+
**Date:** 2026-07-22 (REVISED IN PLACE 2026-07-22 — projection-layer reclassification)
**Landmark Type:** Constant-Triple Convergence — QCalc PROJECTION LAYER (scope corrected)
**Discovery Round:** R365 (`EnhancedBuoyancyQCalcCalculator`)
**Status:** Formal landmark whitepaper — UQFF canonical

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## REVISION NOTICE

Scope corrected per canonical-layer deepsearch (see PAPER_2121 Revision Notice). The R365 triple describes the **QCalc projection layer**. The key reinterpretation: **{$\rho$_vac, c} is not a physics kernel — it is the projection pair.** The wrap buoyancy $Ub = \beta_i \cdot \rho_{vac} \cdot V \cdot c^2$ is the energy-form projection of the canonical quantum chain, whose $\rho_m \rightarrow \rho_E$ conversion is $\rho_E = \rho_m \cdot c^2$ (`uqff_pure_calculator.py` line 33295). The canonical buoyancy $F_{UBi} = (\rho_{SCm} \cdot M/r) \cdot (1+\beta_i \cdot \cos(\pi t_n)+SSq)$ contains **no c and no $\rho$_vac·V·c²**; its kernel is {$\rho_{SCm}$, $\beta_i$}. The prediction-validation record of this paper stands, rescoped as projection-layer statements.

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## Abstract

R365's fill of `EnhancedBuoyancyQCalcCalculator` produced the first single-class occurrence in the R218+ campaign where three independently-derived UQFF constants ($\beta_i$, PAPER_1203; $\rho_{vac}$, PAPER_646/1051; c, PAPER_592) simultaneously appear as class-level primitives in a single wrap closure — the first **Constant-Triple Convergence of the projection layer**. PAPER_2121's forecast (triple by R380) was fulfilled 15 rounds early. $\beta_i$ was simultaneously canonicalized 0.6 → 0.6029 per Rule 2.

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## 1. The Discovery

```python
class EnhancedBuoyancyQCalcCalculator:
BETA_I_PRIMITIVE = 0.6029 # PAPER_1203 canonical
RHO_VAC_PRIMITIVE = _RHO_VAC_UA # PAPER_646/1051: 10·ρ_SCm
```

```
C_PRIMITIVE = 3e8 # PAPER_592

def compute(self, V, SCm, ...):
Ubi = beta_i rho_vac V c2 SCm
```

Three constants, three derivations, one wrap closure. External inputs {V, SCm} are system parametrics.

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## 2. The Projection Identity — why this triple exists

The wrap's structure is now explained by the canonical layer rather than by a kernel postulate:

```
canonical chain: ρ_E = E_total/(c/ω_SCm)³ ; ρ_m = ρ_E/c² (line 33295)
wrap buoyancy: Ub = β_i · (ρ_vac · c²) · V · SCm
■■■ ρ_m→ρ_E conversion = projection of the chain
```

$\rho$**_vac·c² is the energy density the canonical chain assigns to the UA vacuum.** The wrap compresses the canonical F_UBi/F_UBii pair (with its k_spring stiffness and dynamic $\beta$(t,E,Z)) into a single energy-form term. The {$\rho$_vac, c} co-occurrence is therefore **structurally guaranteed in any energy-form wrap** — they are the projection pair, and $\beta$_i rides along as the coupling that survives projection intact (it appears in BOTH layers: canonical F_UBi and the wrap).

$\beta$**_i is the cross-layer invariant of buoyancy.** This is the durable physical content of R365: the same locked $\beta$_i = 0.6029 appears in the canonical kernel {$\rho$_SCm, $\beta$_i} and in the projection triple {$\beta$_i, $\rho$_vac, c}.

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## 3. Prediction Validation (rescoped, record unchanged)

| Landmark | Forecast | Actual | Latency |
|---|:-:|:-:|:-:|
| PAPER_2109 F_TRZ³ 9th | R308-R337 | R309 | 1 round |
| PAPER_2117 quintuplet | R360 | R360 | 0 rounds |
| PAPER_2121 projection triple | R380 | R365 | –15 rounds |

The forecasts were, and remain, statements about which wrap classes carry which constant sets — projection-layer audit predictions, correctly landed.

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## 4. The Canonical Counterpart

The canonical buoyancy machinery this wrap projects (all wired):

```
F_UBi = (ρ_SCm·M/r)·(1 + β_i·cos(πt_n) + SSq) {ρ_SCm, β_i, SSq}
F_UBii = +β(t)·(r/r0)·k_spring·(1+E_n)·|cos(πt_n)| {β_i, k_spring}
k_spring = (ρ_UA/ρ_SCm)·ω_SCm·Φ_res = 1.05e13 {SO_5, ω_SCm, Φ_res}
```

```
β(t,E,Z) = β_i·|E|·Z·cos(πt) DYNAMIC
```

$\omega_{SCm}$ = 1.25 THz and $\Phi_{res}$ = 0.84 — the canonical stiffness constants — appear in **zero** QCalc wrap classes. Their absence from the projection layer is what the original version of this paper failed to notice; their presence in k_spring is where canonical buoyancy actually gets its scale.

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## 5. Gate Assertion

Gate count: 3078 → 3086 (+8 PAPER_2122 assertions, text updated to projection-layer scope).

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## 6. Cross-Paper Links

PAPER_1203 (canonical $\beta_i$ + $F_U$=0 + k_spring), PAPER_646/1051 ($\rho_{vac} = 10\cdot\rho_{SCm}$), PAPER_592 (c), PAPER_1202 (quantum chain), PAPER_1065 (buoyancy variational EOM — canonical Ub derivation chain, wiring queued), PAPER_2121/2123/2124/2125 (revised siblings).

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## 7. Summary Statement (revised)

**PAPER_2122 documents the first projection-layer Constant-Triple Convergence (R365, {$\beta_i$, $\rho_{vac}$, c}) and identifies its structural origin: {$\rho_{vac}$, c} is the projection pair — the wrap's $\rho_{vac}\cdot c^2$ is the canonical quantum chain's mass-to-energy conversion ($\rho_E = \rho_m\cdot c^2$) — while $\beta_i$ is the cross-layer invariant appearing in both the canonical kernel {$\rho_{SCm}$, $\beta_i$} and the wrap. The canonical stiffness constants $\omega_{SCm}$ and $\Phi_{res}$ live exclusively in k_spring on the canonical layer and appear in no wrap. The prediction-validation record (–15 rounds) stands as a projection-layer audit result. $\beta_i$ canonicalized 0.6 → 0.6029 per Rule 2.**

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**Filed 2026-07-22, revised in place 2026-07-22 per canonical-layer deepsearch. Append-only henceforth.**

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# G/c DERIVATION NOTE (appended 2026-07-22, UNIFIED REGISTRY R2 corpus pass)

This paper uses c = 3e8-family literal as published. Per the Unified Registry (R1-adjudicated canonical routes, 2026-07-22):

- **c (speed of light):** canonical route **PAPER_592** — parameter-free $c_{UQFF} = (26\cdot 4\pi/\Phi_{res})\cdot v_F = 2.995\times10$■ m/s (0.13% vs observed; $v_F$ Fermi anchor, c-independent).

Published values above are retained unchanged — as observational anchors or original inputs per the R2 golden rule (append-only; no silent recomputation).
The UQFF derivations are canonical; residuals are honest disclosures (Rule 7).

Registry: UNIFIED_REGISTRY.csv | Program: UNIFIED_REGISTRY_PROGRAM_PLAN.md

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